

**FACULTY OF ENGINEERING AND NATURAL SCIENCES
DEPARTMENT OF BIOENGINEERING**

UNDERGRADUATE THESIS

**In Silico Analysis of the Effects of Non-Canonical Amino Acids
Derived from the Bennu Asteroid on the Structural Stability and
Allosteric Function of the DnaK Chaperone Protein**

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Abstract

In this study, the possible effects of non-canonical amino acids identified in samples obtained from the Bennu asteroid (norvaline [NVA], norleucine [NLE], isoleucine [ILE], valine [VAL], α -aminoisobutyric acid [AIB], α -aminobutyric acid [ABA], and β -alanine [B3A]) on the structural stability and allosteric function of the *E. coli* DnaK chaperone protein were examined using in silico methods [3].

DnaK (HSP70 family) is a molecular chaperone of critical importance for cellular proteostasis that regulates the substrate-binding cycle through its ATP-dependent allosteric mechanism [4]. The study targeted two critical regions: the NBD-SBD linker region (L390–L391) and the substrate-binding domain β -sandwich region (SBD, A429–A435). Free energy changes ($\Delta\Delta G$) of mutant proteins were calculated using the PyRosetta ref2015 energy function [7]; mutant–wild-type superpositions were generated using the Boltz-2 structural prediction model.

The findings revealed that NVA exhibited a strong stabilizing effect in the linker region ($\Delta\Delta G = -6.090$ REU, double mutation L390+L391), while in the SBD region it produced different position-dependent results. VAL displayed a stabilizing effect only at linker L390, while it was found to be destabilizing at other positions. ILE and AIB caused pronounced destabilization at all positions. Calculations could not be performed for NLE, ABA, and B3A due to rotamer library incompatibility or backbone geometry constraints.

1. Introduction

1.1. DnaK and the HSP70 Chaperone Family

Molecular chaperones are fundamental components of cellular proteostasis, mediating the recognition of misfolded or partially unfolded proteins and assisting their return to native conformations [4]. The HSP70 (70 kDa heat shock protein) family, one of the best-characterized of these chaperone families, has a broad evolutionary distribution spanning from prokaryotes to eukaryotes. DnaK, the principal representative of this family in *E. coli*, consists of a 638-amino-acid polypeptide chain and contains two main functional domains.

The N-terminal nucleotide-binding domain (NBD, 1–389) of DnaK catalyzes ATP hydrolysis, while the C-terminal substrate-binding domain (SBD, 393–638) recognizes substrate peptides with hydrophobic cores [11]. The SBD itself consists of two subdomains: the β -sandwich subdomain encloses the substrate peptide, while the α -helical subdomain acts as a lid, regulating substrate binding [1]. DnaK undertakes critical cellular functions such as assisting in the folding of newly synthesized polypeptides, stabilizing damaged proteins under stress conditions, and directing them toward proteolytic degradation pathways [2].

1.2. Allosteric Mechanism and the Importance of the Linker Region

The most notable feature of DnaK is the tight allosteric crosstalk between the NBD and the SBD. In the ATP-bound state, the SBD adopts an open conformation, enabling rapid exchange with substrate; in the ADP-bound or apo state, the SBD lid closes and substrate-binding affinity increases approximately 100–1000-fold [6].

The principal mediator of this allosteric transmission is the linker region (VLLL in *E. coli* DnaK, residues 386–389). The linker functions not merely as a mechanical bridge between the two domains but as an obligatory communication channel for the transmission of the allosteric signal [9]. Mutations in the linker region have been shown to disrupt DnaK's ATPase activity and substrate-binding cycle [5].

1.3. Non-Canonical Amino Acids and the Astrobiological Context

In 2023, analysis of samples returned from the Bennu asteroid as part of JAXA's Hayabusa2 mission identified multiple non-canonical amino acids [3]. This discovery supports the existence of a chemical diversity beyond the standard repertoire of 20 amino acids in the early Solar System. Norvaline (NVA), norleucine (NLE), AIB, and ABA are among these amino acids. The interaction of this broad amino acid pool—delivered to early Earth through asteroid

impacts—with primitive protein systems constitutes an important astrobiological research question [8].

1.4. Research Question and Hypotheses

The central research question of this study is as follows: How do non-canonical amino acids derived from the Bennu asteroid affect the structural stability and allosteric mechanism of the DnaK chaperone? The core hypotheses are: (1) NVA and NLE can be tolerated in the linker region owing to their structural similarity to leucine; (2) the β -branched ILE and the α,α -dialkyl AIB conflict with the narrow hydrophobic geometry of the linker, leading to destabilization; (3) SBD alanine positions are highly sensitive to increases in volume; (4) β -alanine cannot be computed due to backbone incompatibility.

2. Methods

2.1. Protein Structure and Target Positions

The ADP-bound crystal structure of *E. coli* DnaK with PDB code 4B9Q (2.0 Å resolution) was used as the basis for this study [1]. Chain A (587 amino acids) was included in the analysis. Target positions: L390 and L391 in the linker region (PyRosetta numbering: 381–382); A429 and A435 in the SBD β -sandwich region (PyRosetta: 418–424).

2.2. $\Delta\Delta G$ Calculation with PyRosetta

Free energy change calculations were performed using PyRosetta (version 4) with the ref2015 energy function [7]. Protocol: structure loading $\rightarrow$ mutant amino acid placement $\rightarrow \Delta\Delta G = E_{\text{mutant}} - E_{\text{wild-type}}$ calculation. Backbone minimization was not applied (rigid-backbone approach). Rosetta's internal CCD codes were used for non-canonical amino acids (NVL for NVA). Calculations could not be completed for NLE (NLU), ABA, and B3A due to the absence of a valid rotamer library.

2.3. Structural Prediction with Boltz-2

The Boltz-2 model was used via the tamarind.bio platform to generate mutant structures. Input files containing the CCD code and position information were created for each mutant, and a total of 11 structure predictions were completed: 390NVA, 391NVA, 390-391NVA, 390-391ILE, 390-391VAL, 390-391AIB, 429NVA, 429ILE, 435ILE, 429-435ILE, and wild-type.

2.4. PyMOL Visualization

Structural comparisons were performed using PyMOL 2.x. Three viewing angles were recorded for each mutant: close-up view (mutation site), medium view (relevant domain), and distant view (whole protein). Color scheme: blue = wild-type; green = NVA mutant; orange = ILE mutant; cyan = VAL mutant; magenta = AIB mutant.

3. Results

3.1. Linker Region Mutations (L390–L391)

The PyRosetta $\Delta\Delta G$ results for the linker region are summarized in Table 1. Boltz-2 structural images are presented in Figures 1–18.

Table 1

PyRosetta $\Delta\Delta G$ Results for the Linker Region (L390–L391)

Amino Acid	Position	$\Delta\Delta G$ (REU)	Evaluation
NVA	L390	−3.951	Stabilizing
NVA	L391	−3.751	Stabilizing
NVA	L390+L391	−6.090	Stabilizing
VAL	L390	−4.905	Stabilizing
VAL	L391	+2.418	Mildly destabilizing
VAL	L390+L391	−2.403	Stabilizing
ILE	L390	+65.519	Severe destabilization
ILE	L391	+7.318	Destabilizing
ILE	L390+L391	+72.800	Severe destabilization
AIB	L390	+5.700	Destabilizing
AIB	L391	+18.573	Destabilizing
AIB	L390+L391	+25.364	Destabilizing
NLE	L390–L391	ERROR	Rotamer library incompatible
ABA	L390–L391	ERROR	Rotamer library incompatible
B3A	L390–L391	—	Backbone geometry incompatible

Note. Stabilizing ($\Delta\Delta G < 0$); Neutral/mildly destabilizing; Destabilizing ($\Delta\Delta G > 0$); REU: Rosetta Energy Units.

3.1.1. Norvaline (NVA) — Linker

NVA exhibited a pronounced stabilizing effect at both positions in the linker region: L390NVA −3.951 REU, L391NVA −3.751 REU, L390+L391NVA −6.090 REU. The fact that the double mutation produced an approximately cumulative value indicates that the two positions contribute largely independently. In the Boltz-2 structural analyses, the NVA side chain exhibits a hydrophobic contact geometry similar to that of the wild-type leucine, but adopts a more compact conformation due to the absence of one methylene group [1,9].

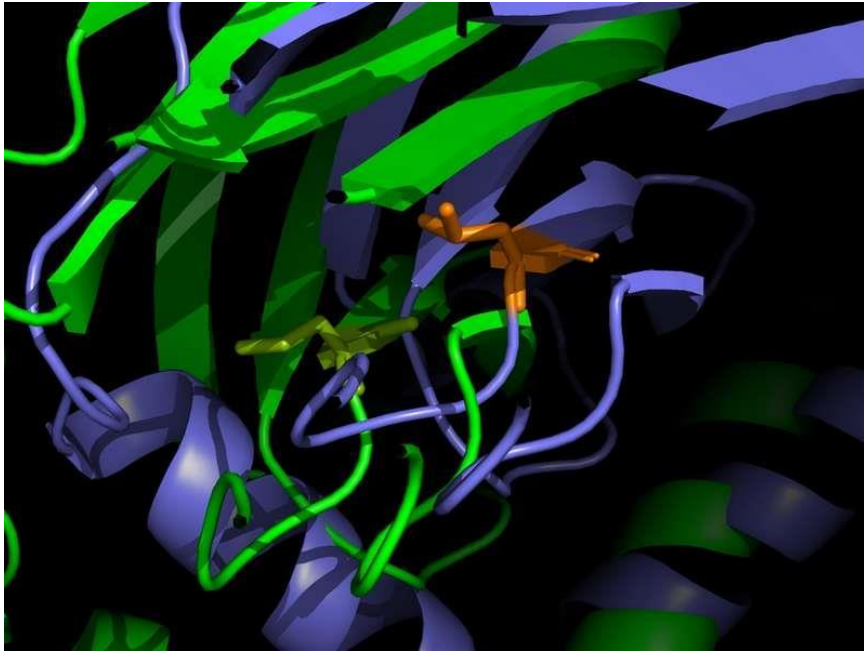


Figure 1. Close-up view of the L390NVA mutant. Blue: wild-type, green: NVA mutant. Yellow: wild-type L390 side chain, green sticks: NVA side chain. The compact placement of NVA within the hydrophobic pocket of the linker is observed.

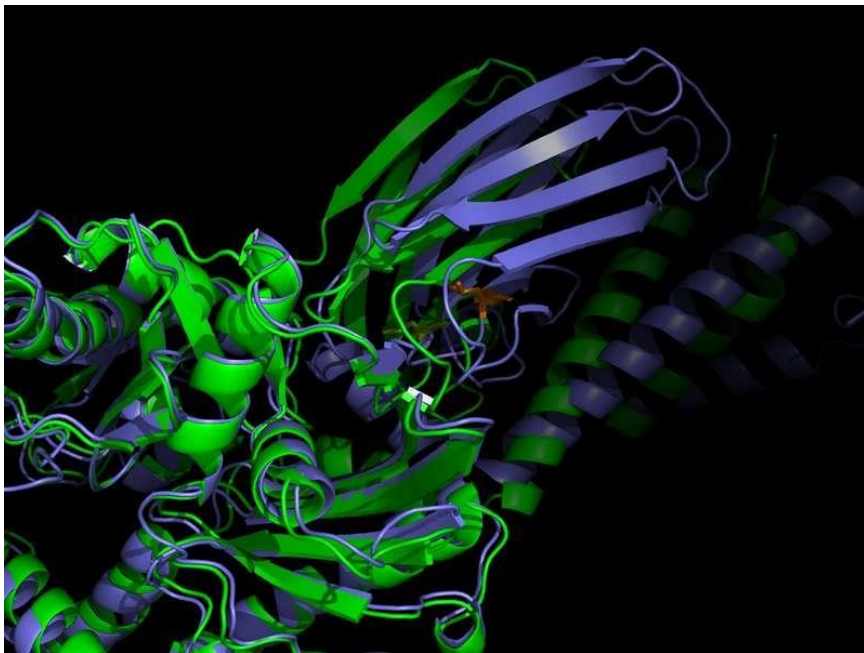


Figure 2. Medium view of the L390NVA mutant. Backbone overlap between wild-type and mutant in the linker and SBD transition region.

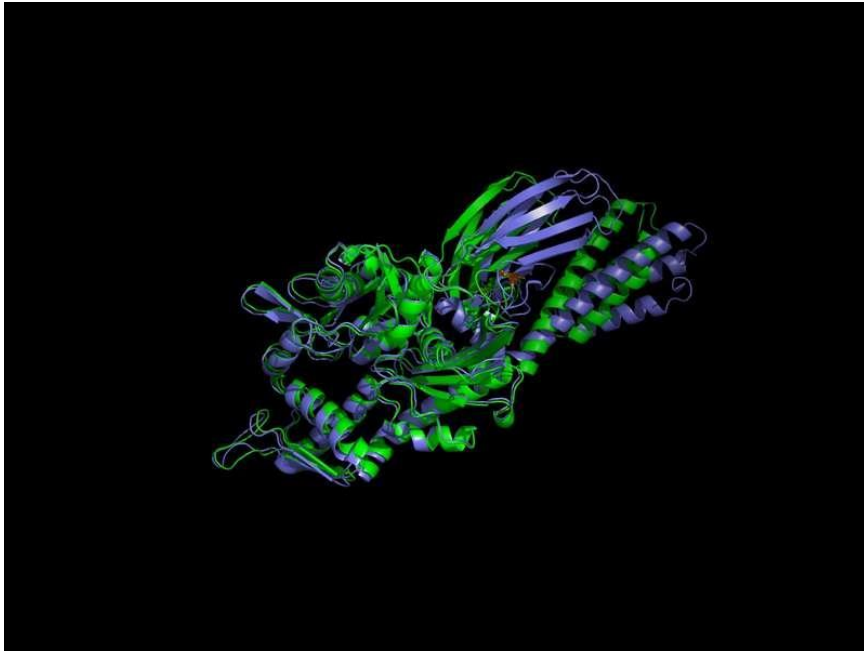


Figure 3. Distant view of the L390NVA mutant. The overall DnaK structure is observed to be preserved.

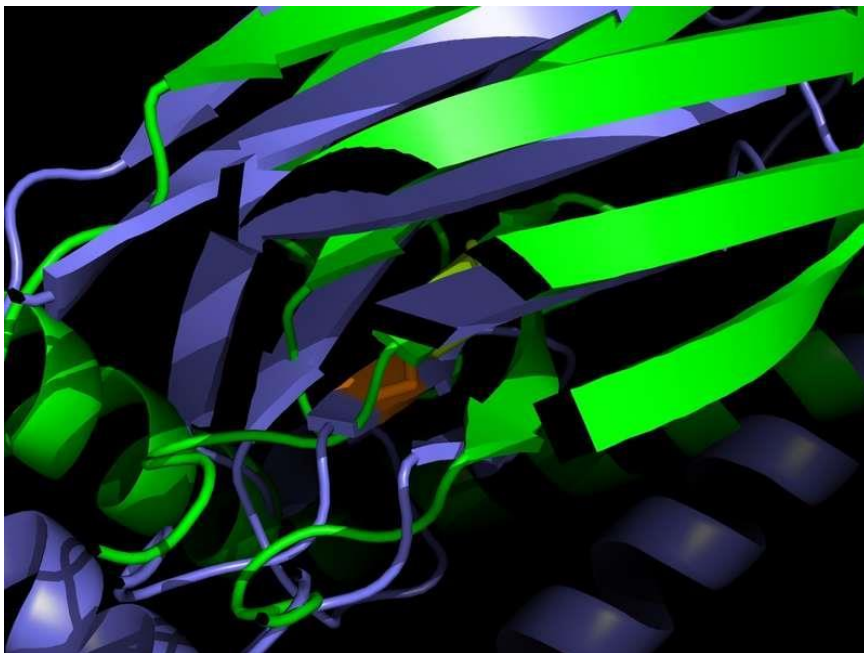


Figure 4. Close-up view of the L391NVA mutant. The conformation of NVA at position L391 and its interaction with surrounding residues.

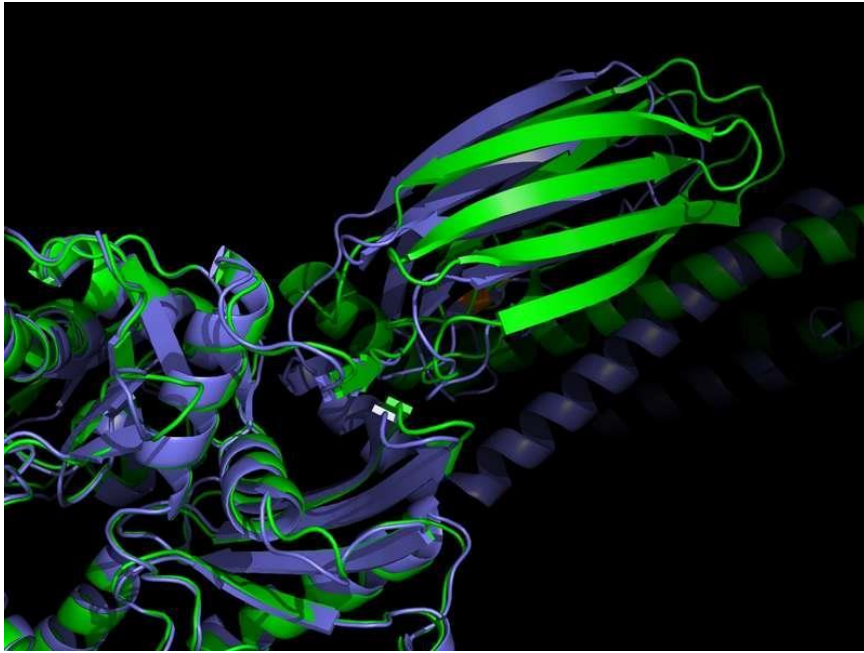


Figure 5. Medium view of the L391NVA mutant. High backbone similarity is observed in the linker region.

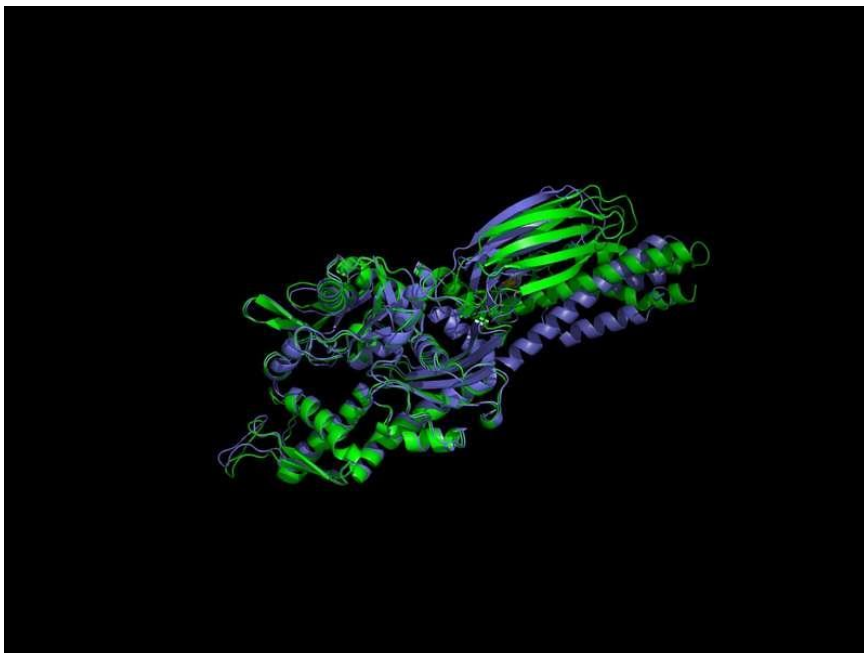


Figure 6. Distant view of the L391NVA mutant. Structural integrity is observed to be preserved at the whole-protein level.

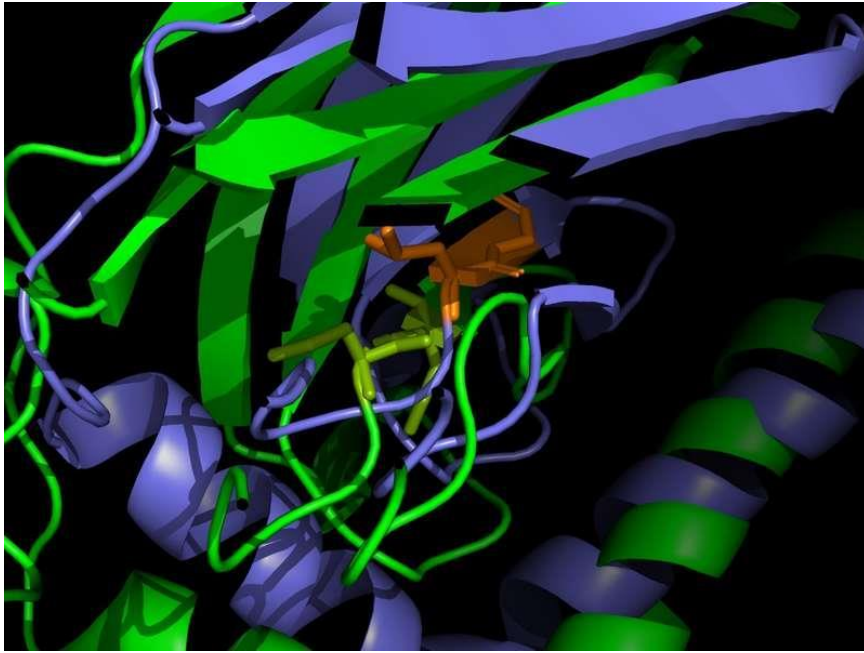


Figure 7. Close-up view of the L390+L391NVA double mutant. The joint conformation of both NVA side chains within the hydrophobic core of the linker.

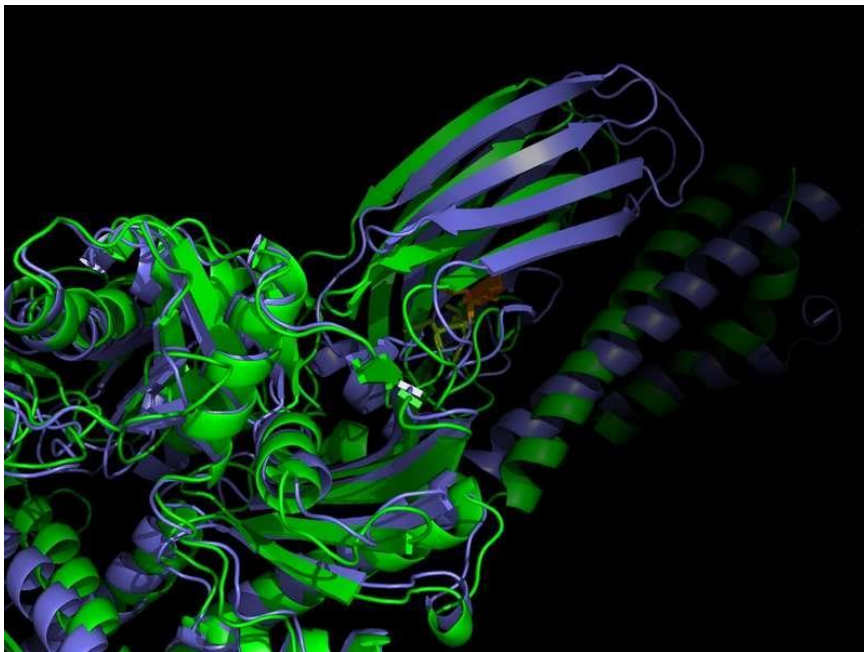


Figure 8. Medium view of the L390+L391NVA double mutant. Preservation of the NBD-SBD connection region despite the double substitution.

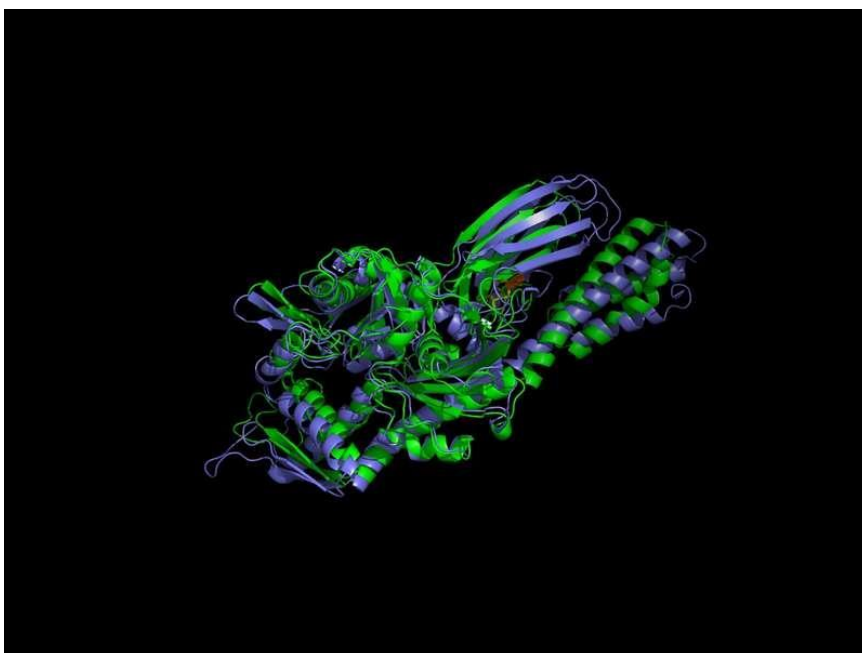


Figure 9. Distant view of the L390+L391NVA double mutant. Overall structure consistent with the -6.090 REU stabilization.

3.1.2. Valine (VAL) — Linker

VAL produced position-dependent different results: L390VAL -4.905 REU (strong stabilization), L391VAL $+2.418$ REU (slight destabilization), L390+L391VAL -2.403 REU (net stabilization; L390 contribution dominant). In the Boltz-2 images, while the overall backbone arrangement is preserved, local conformational variations are present in the linker region [9].

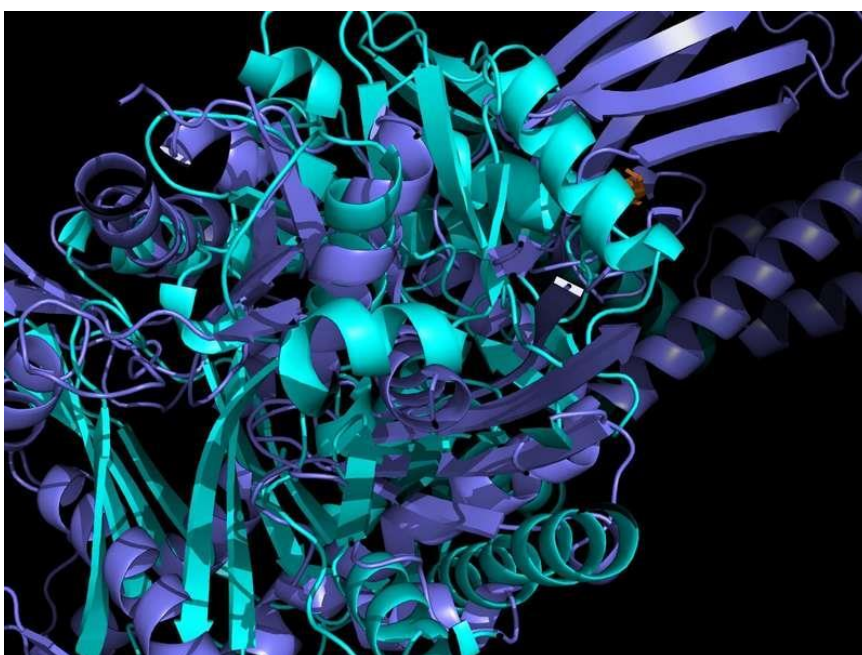


Figure 10. Close-up view of the L390+L391VAL double mutant. Blue: wild-type, cyan: VAL mutant. Placement of the VAL side chains within the linker region.



Figure 11. Medium view of the L390+L391VAL double mutant. Overall backbone preservation in the NBD-SBD transition region.

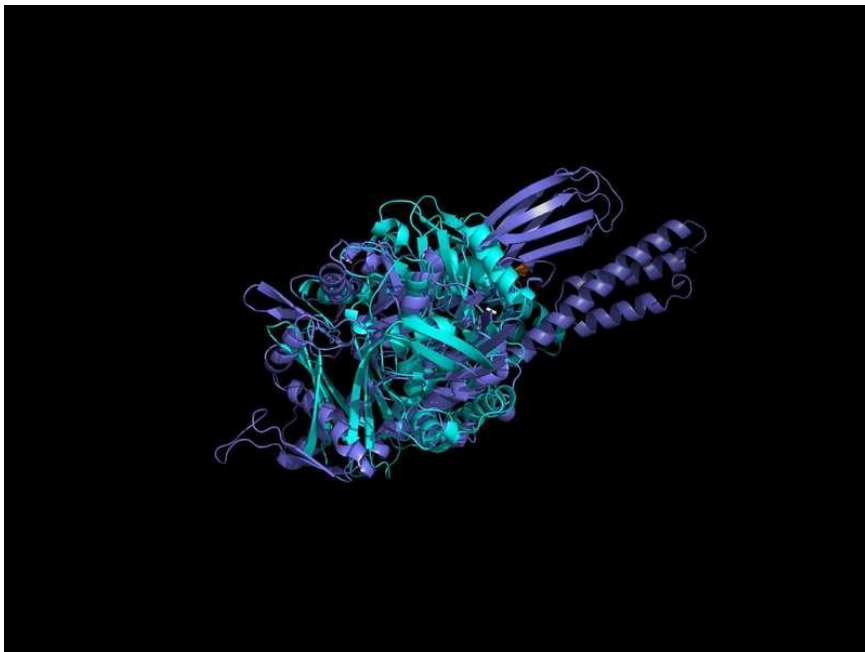


Figure 12. Distant view of the L390+L391VAL double mutant. Overall structural integrity consistent with the net stabilizing $\Delta\Delta G$.

3.1.3. Isoleucine (ILE) — Linker

ILE produced unexpectedly high destabilization values in the linker region: L390ILE +65.519, L391ILE +7.318, L390+L391ILE +72.800 REU. The β -branched structure of ILE (γ -methyl group) creates severe steric clashes with the narrow hydrophobic geometry of the linker. In Boltz-2, the 390-391ILE mutant exhibits pronounced backbone deviations from the wild-type [1,9].

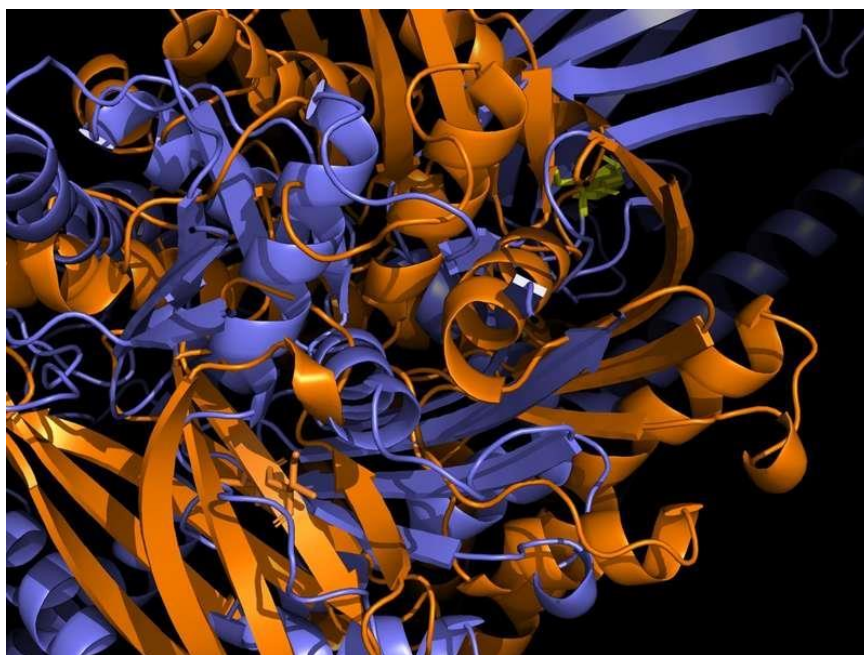


Figure 13. Close-up view of the L390+L391ILE double mutant. Blue: wild-type, orange: ILE mutant. Steric clashes created by the ILE side chains.



Figure 14. Medium view of the L390+L391ILE double mutant. Pronounced backbone deviation from wild-type in the linker region.

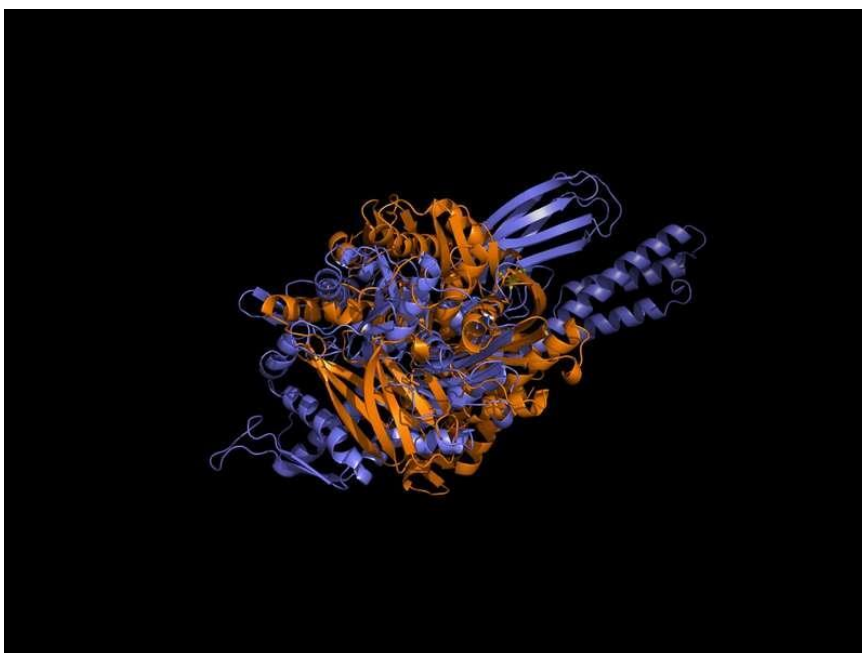


Figure 15. Distant view of the L390+L391IIE double mutant. Overall structural distortion consistent with the high destabilization ($\Delta\Delta G = +72.800$ REU).

3.1.4. α -Aminoisobutyric Acid (AIB) — Linker

AIB was found to be destabilizing at all linker positions: L390AIB +5.700, L391AIB +18.573, L390+L391AIB +25.364 REU. Since the α,α -dialkyl structure of AIB exhibits a strong preference for α -helical conformation [10], its biophysical incompatibility with the flexible structure of the linker is an expected outcome. In Boltz-2, the 390-391AIB mutant shows a rearrangement of the β -strand elements in the linker region.

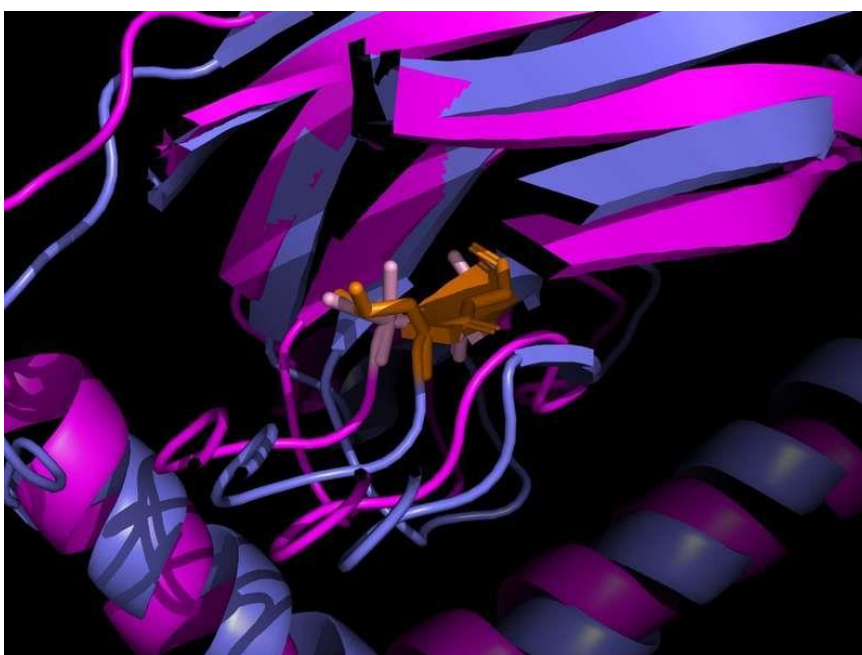


Figure 16. Close-up view of the L390+L391AIB double mutant. Blue: wild-type, magenta: AIB mutant. Steric incompatibility created by the α,α -dialkyl side chain of AIB.

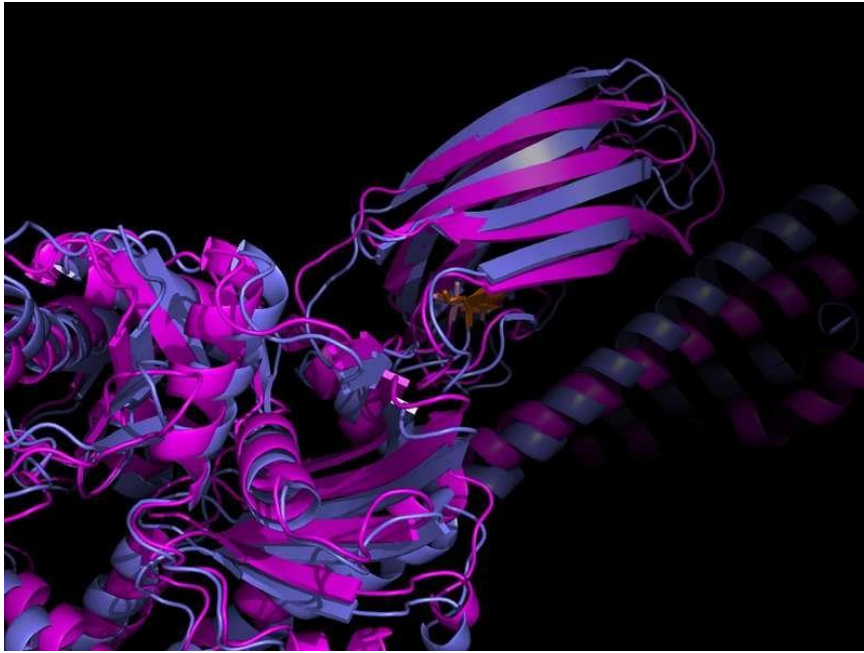


Figure 17. Medium view of the L390+L391AIB double mutant. Notable backbone deviation from wild-type in the linker region.

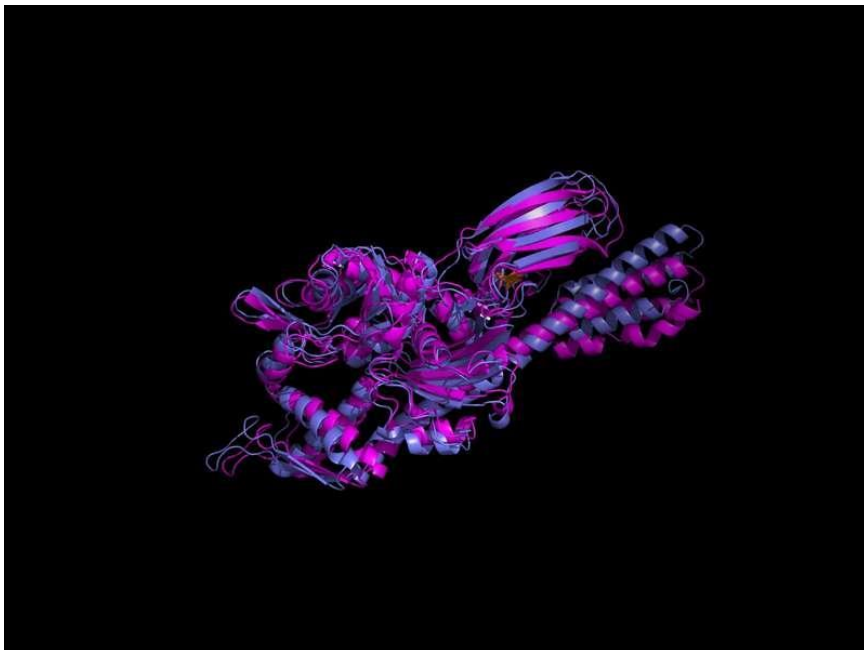


Figure 18. Distant view of the L390+L391AIB double mutant. Overall structural changes consistent with the helical preference of AIB.

3.2. SBD Region Mutations (A429–A435)

The PyRosetta $\Delta\Delta G$ results for the SBD region are summarized in Table 2. Boltz-2 structural images are presented in Figures 19–30.

Table 2

PyRosetta $\Delta\Delta G$ Results for the SBD Region (A429–A435)

Amino Acid	Position	$\Delta\Delta G$ (REU)	Evaluation
NVA	A429	−2.578	Stabilizing
NVA	A435	+5.354	Destabilizing
NVA	A429+A435	+2.776	Mildly destabilizing
VAL	A429	+52.115	Severe destabilization
VAL	A435	+27.256	Severe destabilization
VAL	A429+A435	+79.372	Severe destabilization
ILE	A429	+108.675	Severe destabilization
ILE	A435	+409.291	Severe destabilization
ILE	A429+A435	+517.966	Severe destabilization
AIB	A429	+7.899	Destabilizing
AIB	A435	+113.746	Severe destabilization
AIB	A429+A435	+121.645	Severe destabilization
NLE	A429–A435	ERROR	Rotamer library incompatible
ABA	A429–A435	ERROR	Rotamer library incompatible
B3A	A429–A435	—	Backbone geometry incompatible

Note. Stabilizing ($\Delta\Delta G < 0$); Neutral/mildly destabilizing; Destabilizing ($\Delta\Delta G > 0$); REU: Rosetta Energy Units.

3.2.1. Norvaline (NVA) — SBD

The effect of NVA on the SBD displayed a position-dependent pattern: A429NVA −2.578 REU (stabilizing), A435NVA +5.354 REU (destabilizing), A429+A435NVA +2.776 REU (net destabilization; the adverse contribution of A435 dominant). This asymmetry indicates that A429 and A435 possess different steric environments around the substrate-binding pocket [1,11].

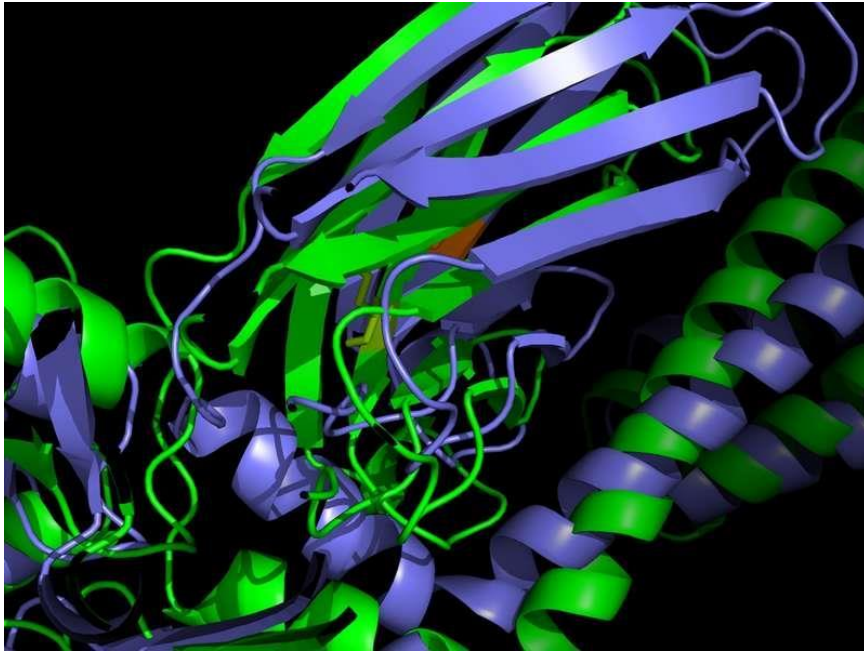


Figure 19. Close-up view of the A429NVA mutant. Blue: wild-type, green: NVA mutant. The conformation of the NVA side chain within the SBD β -sandwich hydrophobic pocket.

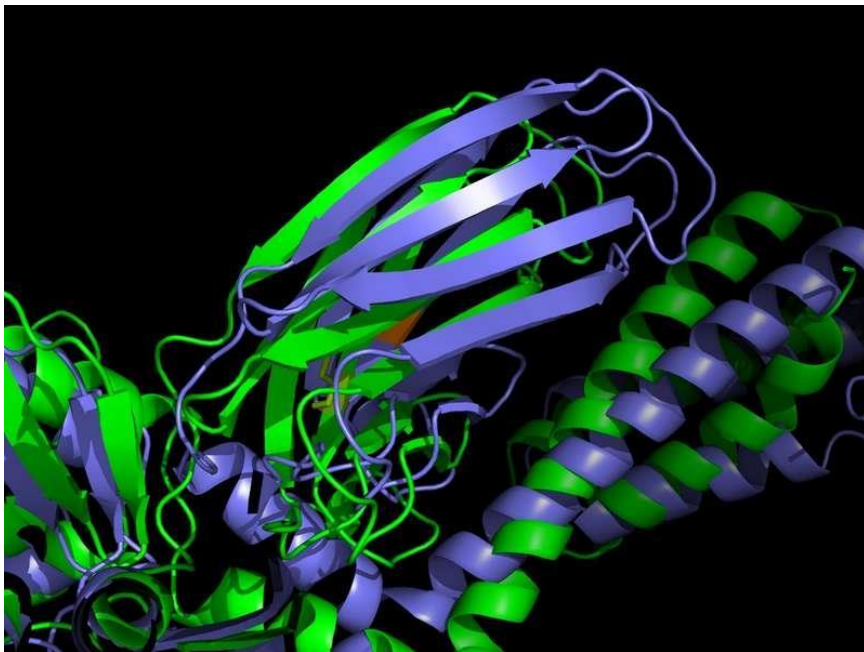


Figure 20. Medium view of the A429NVA mutant. High backbone similarity with wild-type in the SBD region.

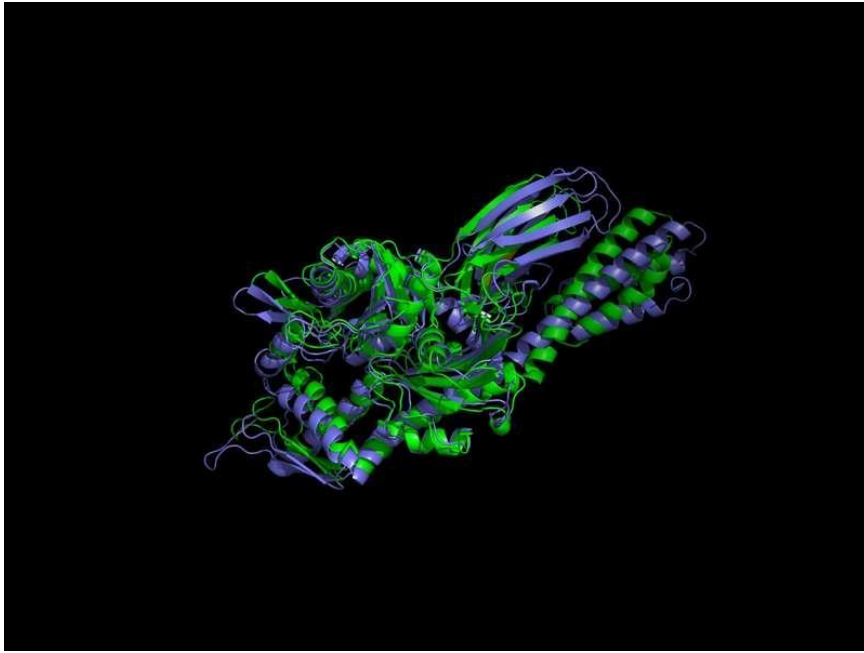


Figure 21. Distant view of the A429NVA mutant. A429NVA (-2.578 REU) is observed to preserve overall structural integrity.

3.2.2. Isoleucine (ILE) — SBD

ILE produced extreme destabilization values in the SBD region: A429ILE $+108.675$, A435ILE $+409.291$, A429+A435ILE $+517.966$ REU. These values reveal that the structural tolerance of the SBD β -sandwich region for bulky amino acids is extremely limited. The existing alanine residues are most likely evolutionarily selected for the substrate-binding geometry; the branched chain of ILE creates extreme steric clashes in this constrained environment [1,11].

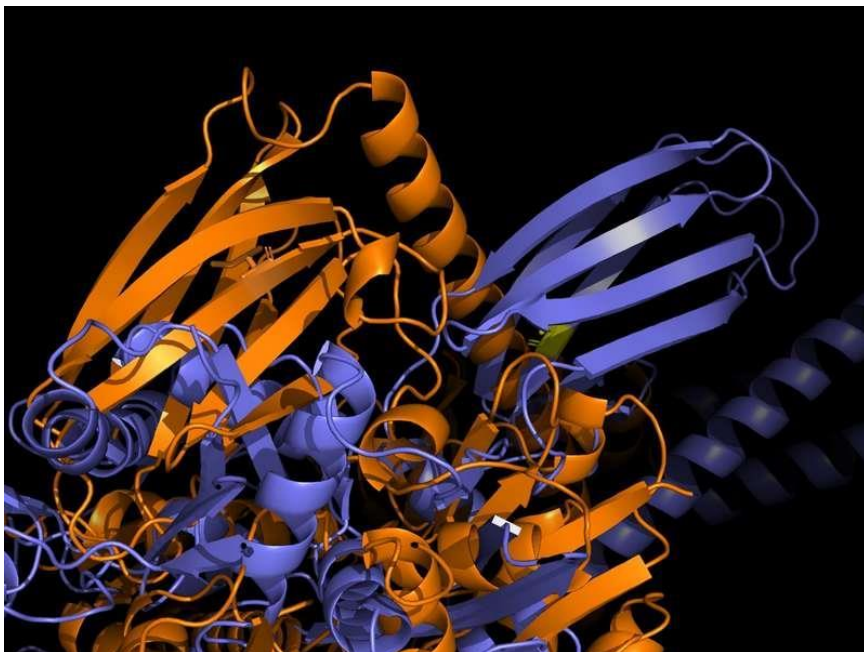


Figure 22. Close-up view of the A429ILE mutant. Blue: wild-type, orange: ILE mutant. Steric clashes created by the ILE side chain within the SBD β -sandwich.

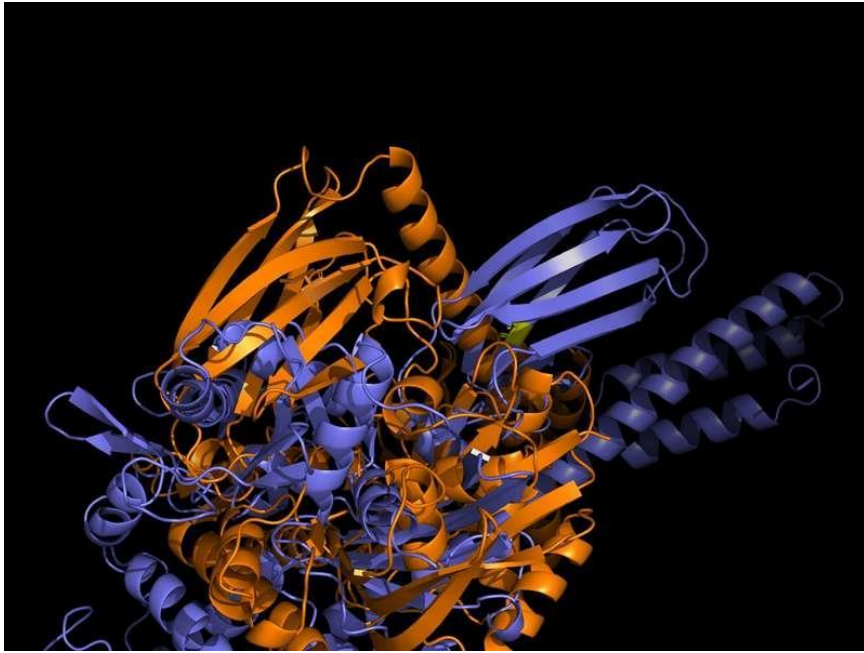


Figure 23. Medium view of the A429ILE mutant. Backbone distortion in the SBD region consistent with +108.675 REU.

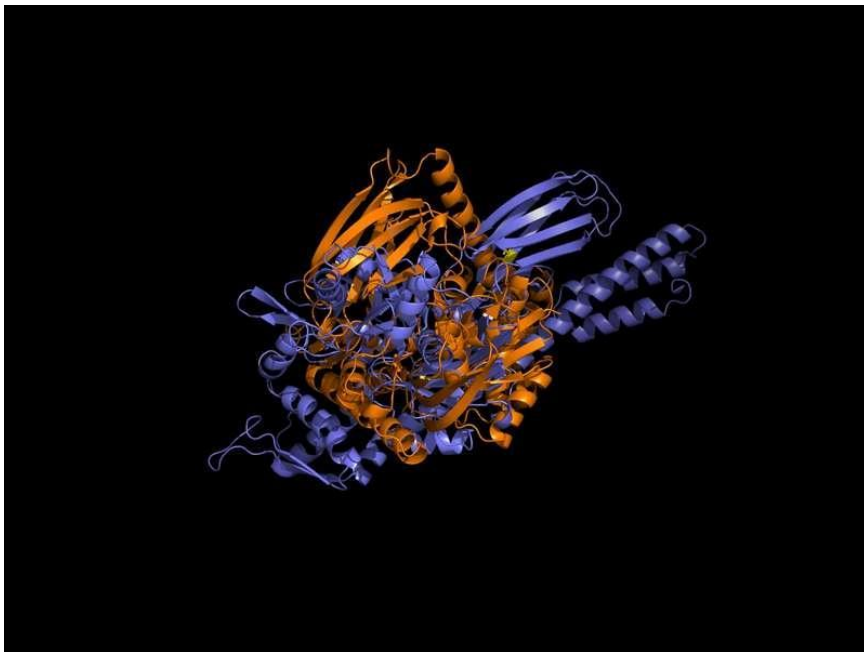


Figure 24. Distant view of the A429ILE mutant. Overall structural deviations consistent with severe destabilization.

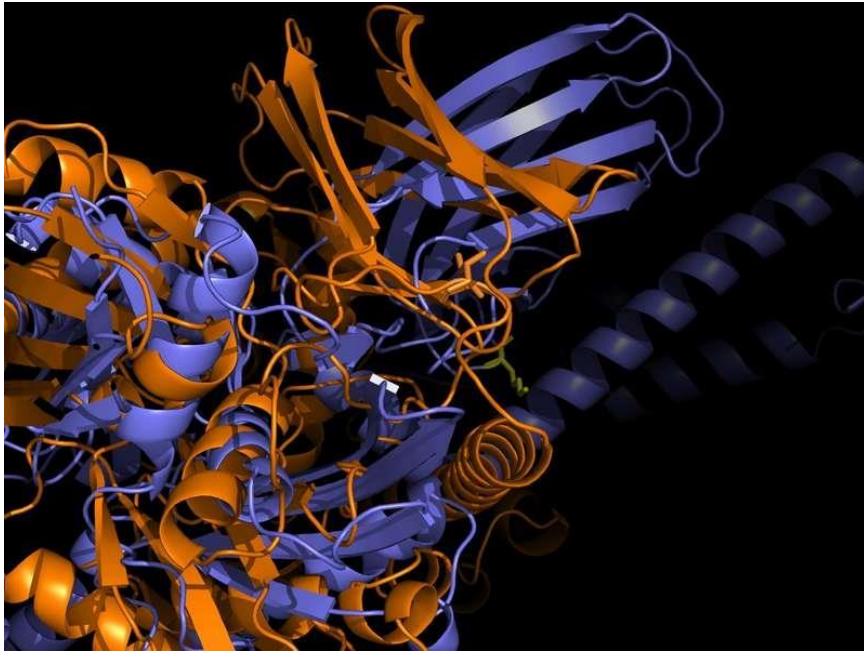


Figure 25. Close-up view of the A435ILE mutant. Steric incompatibility created by ILE at position A435 (+409.291 REU).

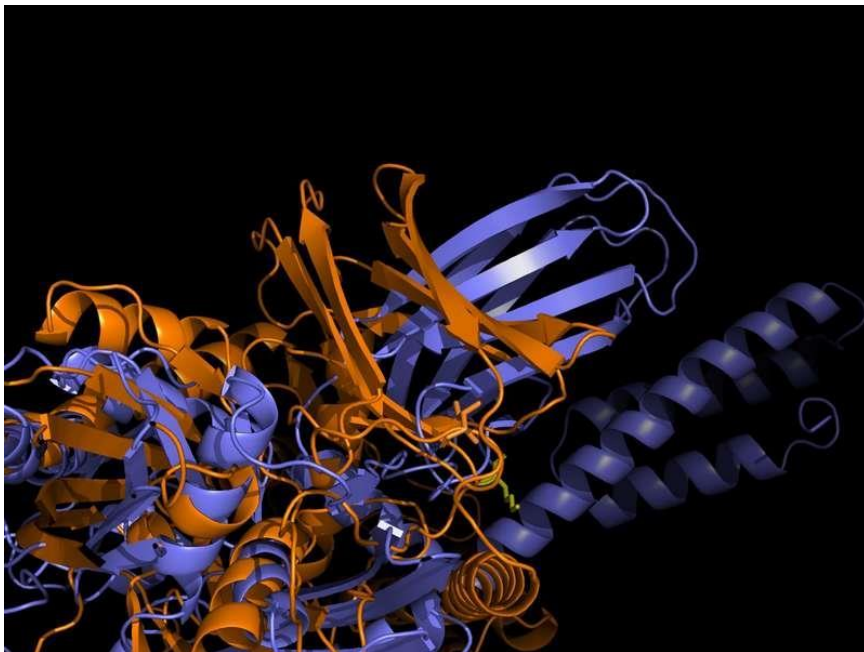


Figure 26. Medium view of the A435ILE mutant. Pronounced SBD backbone deviation from wild-type.

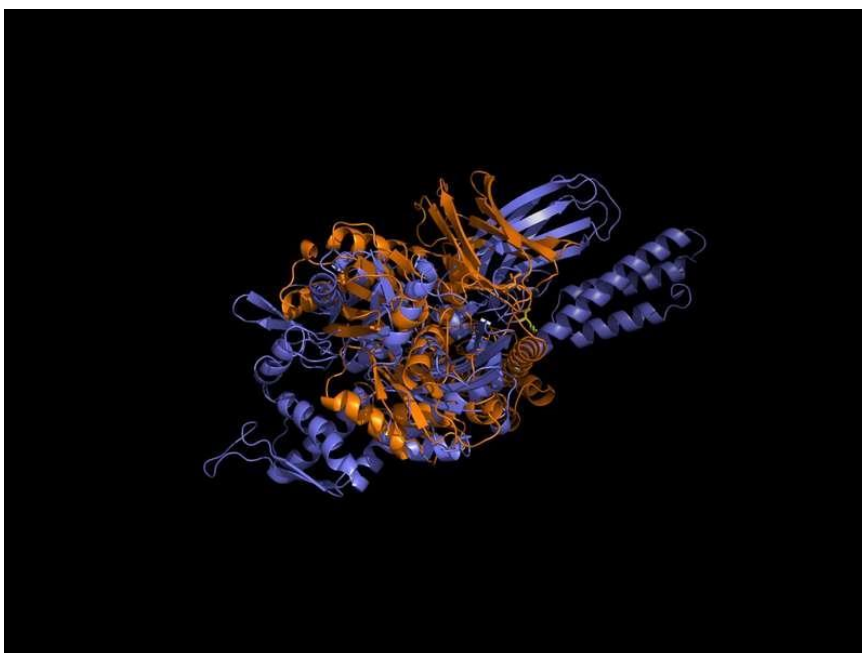


Figure 27. Distant view of the A435ILE mutant. Overall structural disruption consistent with the highest single $\Delta\Delta G$ value.



Figure 28. Close-up view of the A429+A435ILE double mutant. Cumulative steric clashes created by both ILE side chains (+517.966 REU).

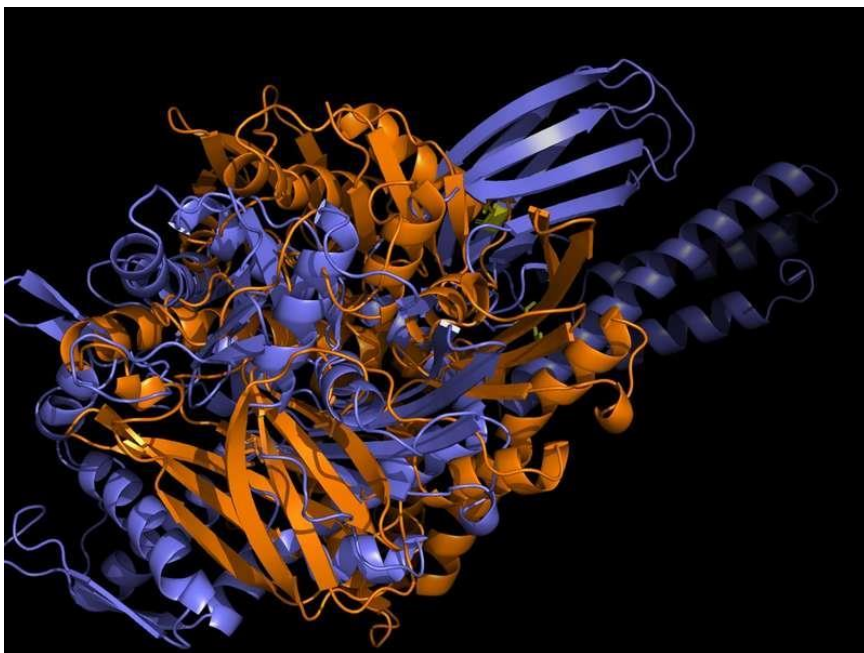


Figure 29. Medium view of the A429+A435ILE double mutant. Pronounced conformational rearrangement in the SBD region.

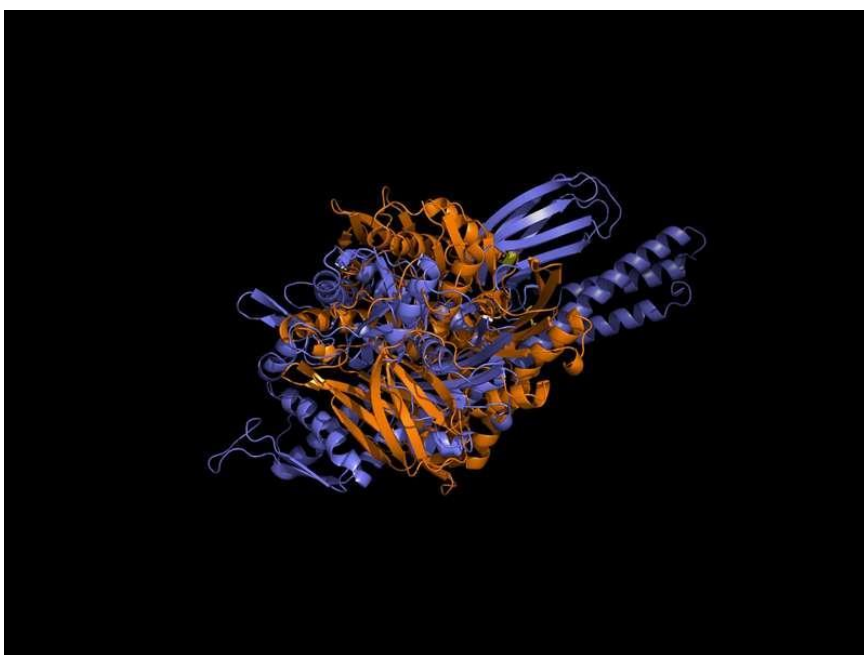


Figure 30. Distant view of the A429+A435ILE double mutant. Structural distortion consistent with the highest total $\Delta\Delta G$ value in the study.

3.2.3. Mutations That Could Not Be Calculated

A PyRosetta rotamer library error was obtained for NLE (NLU code), ABA, and B3A; calculations could not be completed. Since NLE and ABA are not defined within Rosetta's standard parametric framework, only Boltz-2-based structural analysis can be performed for these amino acids. B3A, on the other hand, was excluded from calculation from the outset due to its incompatibility with the standard polypeptide backbone geometry (ϕ/ψ torsion space).

4. Discussion

4.1. The Stabilizing Effect of NVA in the Linker Region

The most notable finding of this study is that NVA can replace leucine residues in the DnaK linker region with $\Delta\Delta G$ values ranging from -3.751 to -6.090 REU. This stabilization may arise from the fact that the straight-chain structure of NVA provides more optimal packing within the hydrophobic pocket of the linker compared to the branched structure of leucine. The confirmation of overall backbone preservation by the Boltz-2 structural analyses indicates that the linker conformational dynamics required for allosteric transmission are largely preserved [6,9].

4.2. Position-Dependent Effect in the SBD Region

The differing effects of NVA at A429 and A435 (-2.578 and $+5.354$ REU) indicate that these positions have distinctly different steric microenvironments. The extremely high destabilization values of VAL and ILE in the SBD demonstrate that the alanine positions are evolutionarily optimized for the substrate-binding geometry; this is also consistent with previous studies showing that even small increases in amino acid volume can lead to severe energy penalties [1,11].

4.3. Astrobiological Interpretation

The findings of this study offer important implications regarding potential protein evolution on early Earth. The structural compatibility of NVA in the linker region and its stabilizing effect support the possibility that this amino acid could have been integrated into primitive chaperone precursors [3,8]. On the other hand, the pronounced destabilization shown by amino acids such as ILE and AIB at certain positions indicates that biophysical constraints were decisive in amino acid selection during early protein evolution.

5. Limitations and Future Work

5.1. Limitations of the Present Study

The absence of backbone minimization in the PyRosetta calculations constitutes the most important methodological limitation; this can lead to overestimations, particularly for high REU values [7]. The lack of a rotamer library for NLE, ABA, and B3A prevents energy-based interpretation for these amino acids. The static structure predictions of Boltz-2 cannot reflect the allosteric dynamics associated with the ATP/ADP cycle. Furthermore, the fact that the remaining 32 Boltz-2 jobs could not be completed due to computational capacity constraints has left some amino acid–position combinations unevaluated.

5.2. Future Work

Repeating the $\Delta\Delta G$ calculations with backbone minimization using the Flex ddG protocol would yield more reliable results. Alternative calculations based on FoldX or Rosetta Relax should be performed for NLE, ABA, and B3A. The allosteric dynamics of stable NVA mutants should be examined using molecular dynamics simulations. Changes in the binding affinity of NVA mutants should be investigated through substrate peptide docking analysis. In the long term, in vitro validation will be provided by non-canonical amino acid incorporation experiments using amber stop-codon suppression.

6. Conclusion

In this study, the effects of seven non-canonical amino acids derived from the Bennu asteroid on the linker (L390–L391) and SBD (A429–A435) regions of the DnaK chaperone protein were evaluated *in silico* using PyRosetta [7] energy calculations and Boltz-2 structural predictions.

Key findings: NVA showed a strong stabilizing effect in the linker region ($\Delta\Delta G = -6.090$ REU, double mutation) and, consistent with the Boltz-2 analyses, preserved the overall backbone geometry while exhibiting local conformational adaptation [1,9]. VAL provided a stabilizing effect at linker L390 while being found destabilizing at other positions. ILE and AIB produced pronounced destabilization at all positions; extreme values reaching 517 REU were obtained for ILE in the SBD. NLE, ABA, and B3A were found to be incalculable.

These findings clearly demonstrate that non-canonical amino acid compatibility varies greatly depending on the structural properties of the amino acid and its position within the protein. The favorable energy profile of NVA in the linker region supports the hypothesis that this amino acid could have been integrated into primitive chaperone precursors if delivered to early Earth via asteroid impact [3,8]. This study presents an interdisciplinary approach that tests an astrobiological hypothesis using *in silico* methods and lays the groundwork for experimental follow-up studies.

References

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